Soft Flight ®
Coating Drum

Generates a gentle folding action that exposes each piece of product to the application of liquid and dry coatings.

Original Instructions

SPRAY DYNAMICS

SOFT FLIGHT® COATING DRUM

CONFIGURATION

This sheet defines the Master Series system for which this manual is supplied. In the event of future needs for parts or service, information on this sheet will assist our Service Technicians and help you get the unit back on-line. Do not remove this sheet from this manual except to make a copy for your service files: then be sure to return this original to the manual.

MASTER SERIES SOFT FLIGHT COATING DRUM

SERIAL NO:	1504
SALES ORDER:	2020053566
MODEL NAME:	COATING DRUM
DRUM TYPE:	SHELL-IN-SHELL
DRUM DIAMETER/LENGTH:	48”Ø X 8’
MOTOR:	1 HP, 208/230-460 VAC, 56C, SS
REDUCER:	10:1, RIGHT HAND, SS
POWER REQUIREMENTS:	480 VAC / 3 PH / 60 HZ
CONTROL PANEL:	32095129 SERIAL #1432

IMPORTANT INFROMATION FOR ORDERING PARTS:

When ordering parts, be sure to list name of part, part number, quantity required, and Machine Serial Number.

SPRAY DYNAMICS

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PREFACE

SAFETY MEASURES

Throughout this manual there are 5 symbols next to information which should be read and carefully understood before doing any maintenance or operating the machine. The symbols are

Electrical warning

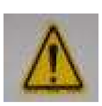
General warning

Hand Entanglement Chain

Hand Entanglement Rotating Gears

Hand Crush and Cut
Force from above

All operators should be adequately trained and specialized personnel should carry out any repairs beyond the regular service of the machine. These would include maintenance personnel, and line managers.

 Keep fingers clear of rotating coating drum.

 Take care when manually moving machine to avoid other personnel and equipment.

 Minimum of 2 persons required to move this machine.

Risk of back injury – use appropriate manual handling techniques.

Keep hands and feet clear of wheels when manually moving machine.

Hazardous Voltages. Before opening any electrical panel, or terminal box, proper Lock-Out/Tag-Out procedures should be followed. Lock-out/Tag-out should also include ANY energy source. These energy sources can include hydraulics, pneumatics, and water. Only competent electrical persons should maintain electrical equipment.

Ear plugs and safety glassed should be worn when in the vicinity of this machine.

Lift equipment in accordance with provided lifting points and/or diagram. Use safe lifting practices. Do not stand under equipment being lifted.

Do not operate without guards in place.

Use care when positioning applicators (liquid and dry) into the coating drum. Repair any sharp edges or burrs to drum cylinder infeed and discharge cones as needed.

Keep hands clear of frame and height adjusting mechanism when manually adjusting drum angle.

The airborne noise levels for the coating drum are dependent upon 2 variables. The speed of the drum and the type of product going through the drum. Most coating drums will be in the range of 60-65 decibels with no product in the drum.

The coating drum is driven by a chain with 2 sprockets. These items are covered by a guard. Do not remove the guard unless proper lock-out/tag-out procedures have been followed for de-energizing the control panel.

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 This symbol is located on the axle guards. These guards protect operators from getting their fingers pinched between the drum and drive wheels. Do not remove the axle guards unless proper lock-out/tag-out procedures have been followed for de-energizing the control panel.

 On some drum cylinders there is an optional clean-out drain plug on the discharge end of the drum. Be aware of this plug any time the coating drum is rotating as the plug could get caught on loose clothing of personnel that get too close.

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GENERAL

The Spray Dynamics Soft Flight Coating Drum® design is a result of over 30 years of experience applying coatings and flavorings onto a wide variety of products and has evolved through a combination of science and art. With each base product being specific to each individual customer, it is important to spend a few minutes understanding the general concept and expectations of the coating drum.

The most important issue to the coating process is the base product feed rate into the coating drum. The base product must be fed into the coating drum at a consistent rate to achieve a uniform coating on the product. After installation, the coating drum is considered "charged with product" when the bed of the base product is uniform in width throughout the entire length of the coating drum. As the coating drum rotates, the base product will travel down the coating drum via a tumbling action. This tumbling action "folds" the product onto itself. The discharge spiral will "dam" the product until there is sufficient product to develop a consistent bed throughout the drum.

Through its tumbling action, the Spray Dynamics' coating drum uniformly exposes each piece of base product to the coating or flavoring section. The coating section (s) can be either liquid or dry, or both. Spray Dynamics offers equipment for both the liquid and dry sections of the coating drum. Spray Dynamics' equipment often is custom designed around the customer's base product and process parameters.

Please refer to the **OPERATION** section for more information.

Spray Dynamics' coating drum standard features include; stainless steel coating drum and frame, adjustable tilt angle, and variable-speed drive on the coating drum rotation.

This equipment is not intended to coat any product outside the scope of the proposal.

Any variations in the design of this system will void any implied warranty, unless authorized by Spray Dynamics. Any new products to be coated must be assessed for the potential risks and hazards and suitable measures must be taken.

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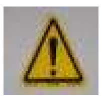
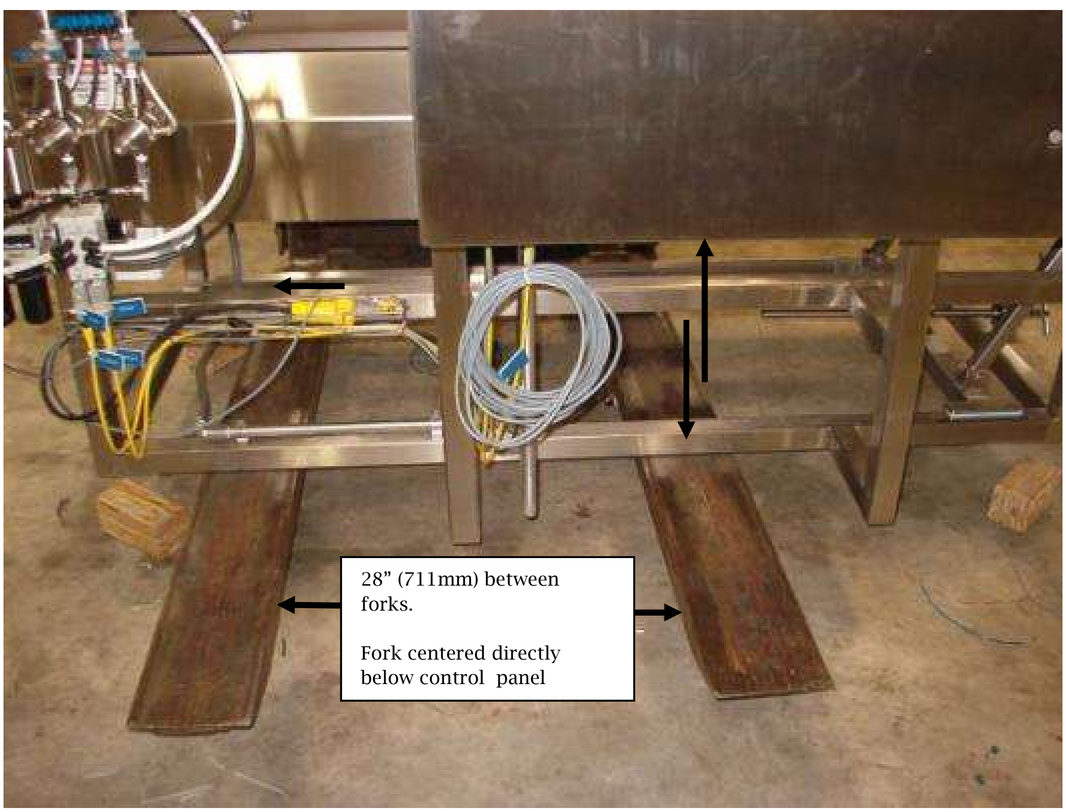
INSTALLATION

 Lift equipment in accordance with provided lifting points and/or diagram. Use safe lifting practices. Do not stand under equipment being lifted.

 It is recommended that 3 people install the components of this machine at its final location. Enough space must be given to allow for installation and removal of the coating drum for cleaning, sanitation, and maintenance.

 Place forks in the center of the coating drum frame for best stability while moving to final location or for raising high enough to insert castors on initial setup. See picture below.



Once the drum is received it may be required to place the drum shell onto the drum frame. Please follow the directions for lifting and moving the drum shell using the yellow lifting straps provided. The drum shell is designed to be placed on the drum frame in one position only. This is determined by the drum rotation and the design of the flights inside of the drum. Refer to the general assembly drawing for proper placement of drum cylinder onto frame. If the drum cylinder arrived already on its frame, you must remove the wood inserts between the drum cylinder and frame so that the cylinder rests on the 4 red drive wheels.

Cut and/or remove all packaging material from the coating drum. This includes banding material and wood pieces.

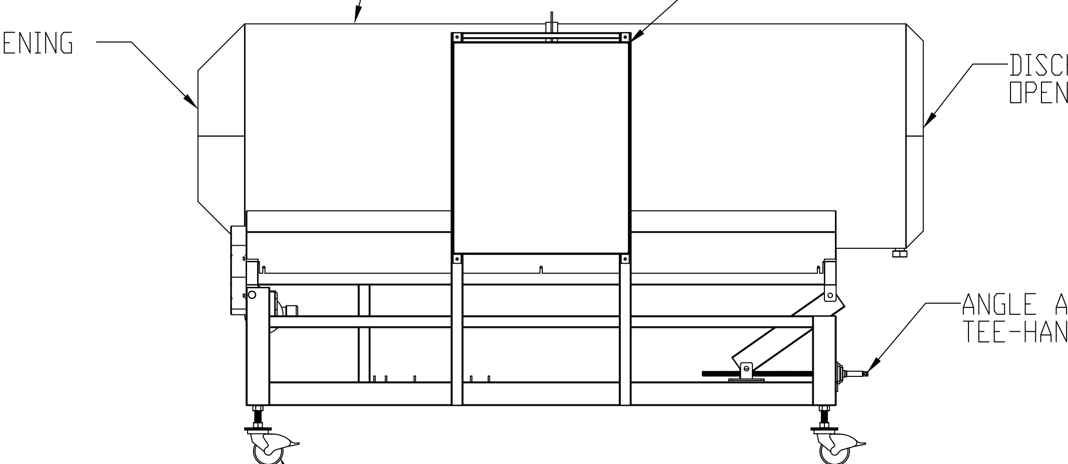
In addition to the Drum Shell, the optional adjustable swivel casters may need to be inserted into the bottom of the drum frame. These get screwed into the drum frame to the desired height. This height is determined after the drum is in place and the proper drum angle has been determined for the process. A good starting point for the drum angle is 1° to 2° from horizontal.

Please be sure to look at the assembly drawings accompanying this manual for the proper set up and orientation and electrical hookup.

SPRAY DYNAMICS

SOFT FLIGHT® COATING DRUM

Pictured is a typical Spray Dynamics Coating Drum.



CONTROLS

 The Spray Dynamics' control panel requires a 480-volt, 60 Hz, 3 phase power drop. The motor supplied on the coating drum is a 208/240-480 VAC wash down motor. The power is converted to a variable Hertz AC source within the NEMA 4 control box. Please refer to the Equipment Drawings.

CONTROL BOX PANEL DOOR

The control panel operating controls are as follows. **NOTE: The line power is live within the control box even when the coating drum is not rotating. Always unplug line power cord or disconnect the power before opening the front cover of the control box. Follow proper Lock-out/Tag-out procedures**

On button: Sends a signal to the VFD, starting the motor, therefore, causing drum rotation.

Off button: Sends a signal to the VFD, stopping the motor, causing the drum to stop rotating.

Speed Control: Controls drum rotation speed. The majority of base products perform the best when the coating drum rotates between four (4) rpm and fourteen (14) rpm. Once the optimal coating drum rotation is determined, the scale should be used for repeatability.

E-Stop: Red System stop button for immediate system shutdown.

***** WARNING *****

 **All controls inside the control console have been factory set for optimum performance. Recalibration, alteration or substitution of any of these components without specific instructions from the factory may result in dangerous operating Conditions and/or failure and possible damage to the equipment. Any modification will invalidate the warranty.**

SPRAY DYNAMICS

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REGULAR CLEANING

1. Remove Product from Drum
2. Note or mark position of drum angle.
3. Flatten out drum angle to horizontal'
4. Fill drum with hot soapy water then turn on drum and set to around 4-6 rpm. Let the drum rotate for several minutes.
5. Stop drum and use soft bristle brush to aid in removal of any product or material stuck to the drum.
6. Once the drum is clean reverse the rotation of the drum to drain using the spiral discharge.
7. At this point you can use a hose to rinse out the drum or you can fill the drum with clean water and repeat steps 4-6.
8. Blow Excess Water OFF with air nozzle

NOTE: Control boxes are wash down, **NOT** high-pressure hose down.

DO NOT use High Pressure Hose on Control Boxes or solenoid valves.

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OPERATION

As discussed in the GENERAL section, a few moments spent in adjusting the machine before starting production should result in a consistent tumbling action and a well-coated product. Product transfers down the drum via a gentle folding action to the discharge spiral. The discharge spiral dams the product and provides further mixing until the product is discharged from the drum. Coating drum speed varies with the different base products. The typical drum rotation usually falls between six (6) rpm and fourteen (14) rpm.

 Spray Dynamics' coating drums are equipped with an adjustable lift or angle control located at one end of the frame. The coating drum angle should be such that at production rates, there is a uniform bed of product along the entire length of the drum. At the proper angle, the product bed should neither taper, nor fan-out near the discharge end of the drum. Typical angles at production rates usually fall around the one and one half degree (1-1/2°) range. If the coating drum angle is too steep, product will travel quickly through the coating drum. At steep angles, the base product coating may be uneven and/or insufficient. If the angle is too shallow, the base product will spend an excessive amount of time in the coating area and/or the base product will spill out the in-feed end of the drum.

SPRAY DYNAMICS

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MAINTENANCE

Hazardous Voltages. Before opening any electrical panel, or terminal box, proper Lock-Out/Tag-Out procedures should be followed. Lock-out/Tag-out should also include ANY energy source. These energy sources can include hydraulics, pneumatics, and water. Only competent electrical persons should maintain electrical equipment.

Do not operate without guards in place.

Lock-Out/ Tag-Out before performing maintenance on the machine.

Keep fingers clear of rotating coating drum.

Take care when manually moving machine to avoid other personnel and equipment.

Keep hands and feet clear of wheels when manually moving machine.

Minimum of 2 persons required to move this machine.

Risk of back injury – use appropriate manual handling techniques.

Check the condition and security of the static earth discharge strap (grounding brush) between the drum and base frame once every 3 months.

Check to ensure setscrews are tight in axle bearings monthly.

Use care when positioning applicators (liquid and dry) into the coating drum. Repair any sharp edges or burrs to drum cylinder infeed and discharge cones as needed.

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Standard Spray Dynamics coating drums have pillow block bearings, a flange bearing on the adjustable lift, a gear reducer and a chain drive from the gear reducer to the axle. Although all these items are lubricated prior to shipping, the following guidelines should be followed for optimum performance.

Pillow Block & Flange Bearing: Lubricate once per week with food grade grease.

Chain: Lubricate once per month with gear lube oil.

Gear Reducer: Lubricated for life by the manufacturer but oil level should be checked once per month.

If these guidelines are adhered to, your Spray Dynamics Soft Flight coating drum will give you years of service. If you ever have any questions, please contact us at:

SANITATION

Machinery should be cleaned and sanitized on a regular schedule to prevent any microbial formations. Each facility should have its own schedule for cleaning and products used for this purpose. ZEP Superior Solutions has several products used for cleaning, degreasing, and sanitizing equipment in the food industry with offices in the United States, France, Italy, Holland, and the United Kingdom.


Cleaning and sanitation should occur daily or more often as your facility may require.

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RECOMMENDED SPARE PARTS LIST

PART #	QTY	DESCRIPTION
20563616	1	Motor, 1HP, 208/230-460VAC, 56C, SS
20563155	1*	Coating Drum Guide Roller
20572344	1	10:1 Gear Reducer, 56C, SS
20564082	4	Bearing, Pillow Block, SS, 1-1/4”
20563445	1	4” X 6” Dia. Drum Wheel, 1-1/4” Bore

***HIGHLY RECOMMENDED TO HAVE ON HAND**

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